

Question	Answer	Marks
1(a)(i)	distance in a specified direction (from a point)	B1
1(a)(ii)	change in velocity / time (taken)	B1
1(b)(i)	vertical component of velocity = $(5.5^2 - 4.6^2)^{1/2} = 3.0 \text{ (ms}^{-1}\text{)}$ or $5.5 \cos \theta = 4.6$ (so $\theta = 33.2^\circ$) and $5.5 \sin 33.2^\circ = 3.0 \text{ (ms}^{-1}\text{)}$	A1
1(b)(ii)	$s = ut + \frac{1}{2}at^2$ $0 = (3.0 \times t) - (\frac{1}{2} \times 9.81 \times t^2)$ or $v = u + at$ $-3.0 = 3.0 - 9.81t$	C1
	$t = 0.61 \text{ s}$	A1
1(b)(iii)	$d = 4.6 \times 0.61$ $= 2.8 \text{ m}$	A1
1(b)(iv)	$E = \frac{1}{2}mv^2$	C1
	ratio = $(\frac{1}{2} \times m \times 4.6^2) / (\frac{1}{2} \times m \times 5.5^2)$ or ratio = $(\frac{1}{2} \times m \times 5.5^2 - m \times 9.81 \times 0.459) / (\frac{1}{2} \times m \times 5.5^2)$	C1
	ratio = 0.70	A1
1(c)	straight line from positive value of v_y at $t = 0$ to negative value of v_y	M1
	straight line ends at $t = T$ and final magnitude of v_y greater than initial magnitude of v_y	A1

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2(c)	(i) 10 15 20 25 30	01